EXT 2: Complex Numbers (Ext2), N2 Using Complex Numbers (Ext2)

Geometrical Implications of Complex Numbers

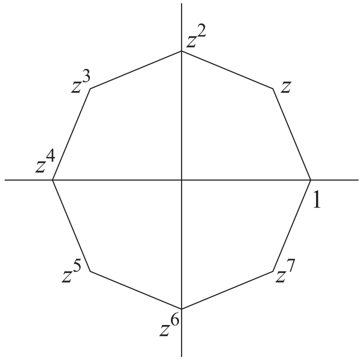
Teacher: Nardene Dodd

Exam Equivalent Time: 108 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*

Questions

1. Complex Numbers, EXT2 N2 2017 HSC 1 MC

The complex number z is chosen so that $1, z, \ldots, z^7$ form the vertices of the regular polygon shown.



Which polynomial equation has all of these complex numbers as roots?

A. $x^7 - 1 = 0$

B. $x^7 + 1 = 0$

C. $x^8 - 1 = 0$

D. $x^8 + 1 = 0$

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2. Complex Numbers, EXT2 N2 2017 HSC 3 MC

Which complex number lies in the region $2 < |z - 1| < 3$?

A. $1 + \sqrt{3}i$

B. $1 + 3i$

C. $2 + i$

D. $3 - i$

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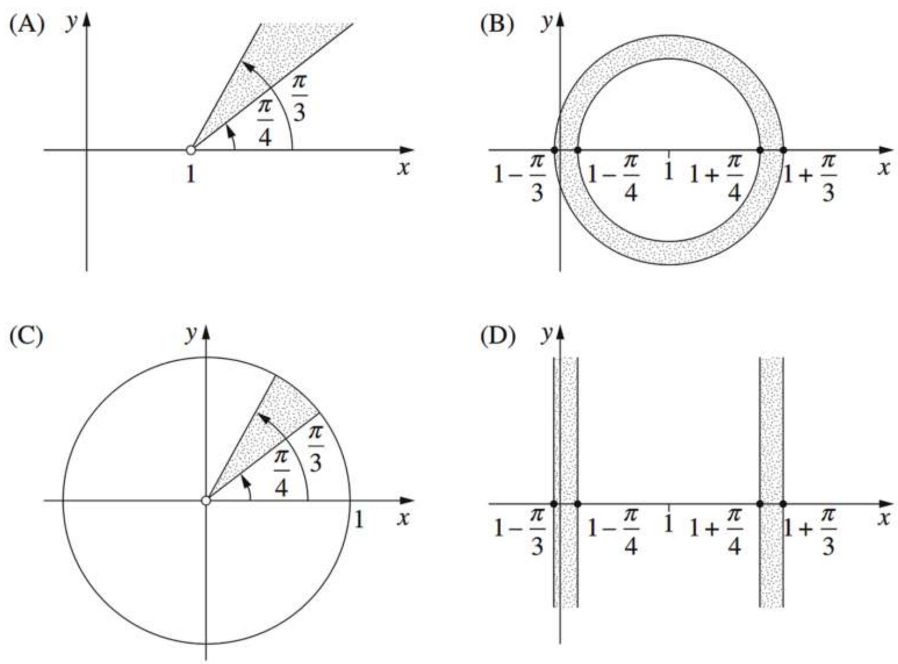
3. Complex Numbers, EXT2 N2 2013 HSC 5 MC

Which region on the Argand diagram is defined by $\dfrac{\pi}{4} \leq |z - 1| \leq \dfrac{\pi}{3}$?



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4. Complex Numbers, EXT2 N2 2015 HSC 9 MC

The complex number z satisfies $|z - 1| = 1$.

What is the greatest distance that z can be from the point i on the Argand diagram?

A. 1

B. $\sqrt{5}$

C. $2\sqrt{2}$

D. $\sqrt{2} + 1$

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5. Complex Numbers, EXT2 N2 2016 HSC 5 MC

Multiplying a non-zero complex number by $\dfrac{1-i}{1+i}$ results in a rotation about the origin on an Argand diagram.

What is the rotation?

A. Clockwise by $\dfrac{\pi}{4}$

B. Clockwise by $\dfrac{\pi}{2}$

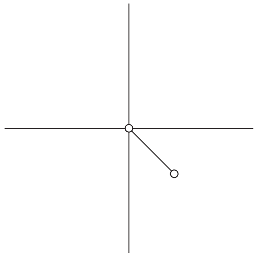
C. Anticlockwise by $\dfrac{\pi}{4}$

D. Anticlockwise by $\dfrac{\pi}{2}$

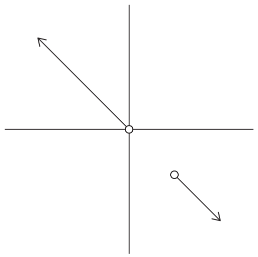
6. Complex Numbers, EXT2 N2 2018 HSC 7 MC

Which diagram best represents the solutions to the equation $\arg(z) = \arg(z+1-i)$?

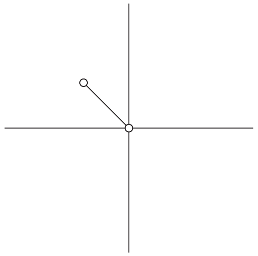
A.



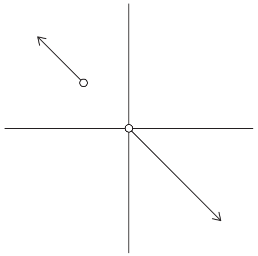
B.



C.

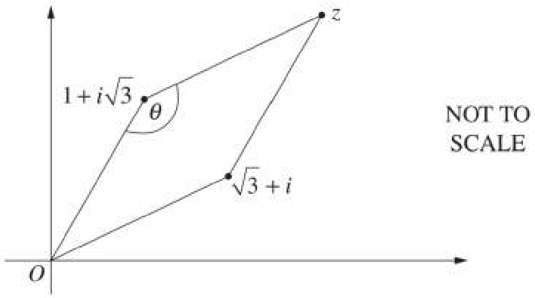


D.



7. Complex Numbers, EXT2 N2 2011 HSC 2b

On the Argand diagram, the complex numbers $0, 1+i\sqrt{3}, \sqrt{3}+i$ and z form a rhombus.



i. Find z in the form $a+ib$, where a and b are real numbers. *(1 mark)*

ii. An interior angle, θ, of the rhombus is marked on the diagram.
Find the value of θ. *(2 marks)*

8. Complex Numbers, EXT2 N2 2009 HSC 2d

Sketch the region in the complex plane where the inequalities $|z-1| \leq 2$ and $-\dfrac{\pi}{4} \leq \arg(z-1) \leq \dfrac{\pi}{4}$ hold simultaneously. *(2 marks)*

9. Complex Numbers, EXT2 N2 2010 HSC 2c

Sketch the region in the complex plane where the inequalities $1 \leq |z| \leq 2$ and $0 \leq z+\bar{z} \leq 3$ hold simultaneously. *(2 marks)*

10. Complex Numbers, EXT2 N2 2012 HSC 11b

Shade the region on the Argand diagram where the two inequalities

$|z+2| \geq 2$ and $|z-i| \leq 1$

both hold. *(2 marks)*

11. Complex Numbers, EXT2 N2 2017 HSC 11c

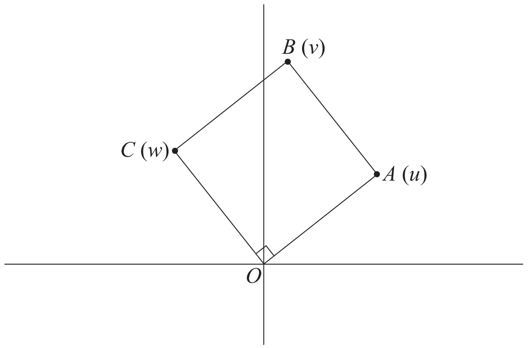
Sketch the region in the Argand diagram where

$-\dfrac{\pi}{4} \leq \arg \leq 0$ and $|z-1+i| \leq 1$. *(2 marks)*

12. Complex Numbers, EXT2 N2 2018 HSC 11d

The points A, B and C on the Argand diagram represent the complex numbers u, v and w respectively.

The points O, A, B and C form a square as shown on the diagram.



It is given that $u = 5 + 2i$.

i. Find w. *(1 mark)*

ii. Find v. *(1 mark)*

iii. Find $\arg\left(\dfrac{w}{v}\right)$. *(1 mark)*

13. Complex Numbers, EXT2 N2 2007 HSC 2c

The point P on the Argand diagram represents the complex number z, where z satisfies

$$\frac{1}{z} + \frac{1}{\overline{z}} = 1.$$

Give a geometrical description of the locus of P as z varies. *(3 marks)*

14. Complex Numbers, EXT2 N2 2013 HSC 11e

Sketch the region on the Argand diagram defined by $z^2 + \overline{z}^2 \le 8$. *(3 marks)*

15. Complex Numbers, EXT2 N2 2014 HSC 11c

Sketch the region in the Argand diagram where $|\,z\,| \le |\,z-2\,|$ and $-\dfrac{\pi}{4} \le \arg z \le \dfrac{\pi}{4}$. *(3 marks)*

16. Complex Numbers, EXT2 N2 2019 HSC 12a

Sketch the region defined by $\dfrac{\pi}{4} \le \arg(z) \le \dfrac{\pi}{2}$ and $\mathrm{Im}(z) \le 1$. *(2 marks)*

17. Complex Numbers, EXT2 N2 2004 HSC 2c

Sketch the region in the complex plane where the inequalities

$|z + \overline{z}| \le 1$ and $|z + i| \le 1$

hold simultaneously. *(3 marks)*

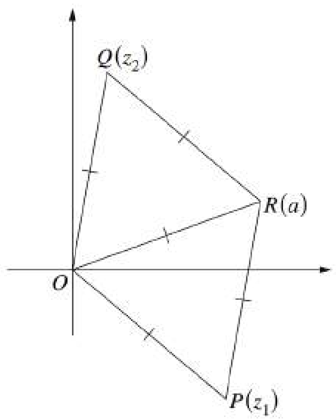
18. Complex Numbers, EXT2 N2 2005 HSC 2c

Sketch the region on the Argand diagram where the inequalities

$|z - \overline{z}| < 2$ and $|z - 1| \ge 1$

hold simultaneously. *(3 marks)*

19. Complex Numbers, EXT2 N2 2007 HSC 2d



The points P, Q and R on the Argand diagram represent the complex numbers z_1, z_2 and a respectively.

The triangles OPR and OQR are equilateral with unit sides, so $|\,z_1\,| = |\,z_2\,| = |\,a\,| = 1$.

Let $\omega = \cos\dfrac{\pi}{3} + i\sin\dfrac{\pi}{3}$.

i. Explain why $z_2 = \omega a$. *(1 mark)*

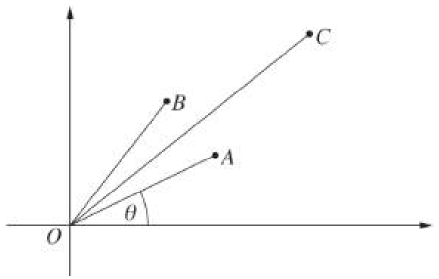
ii. Show that $z_1 z_2 = a^2$. *(1 mark)*

iii. Show that z_1 and z_2 are the roots of $z^2 - az + a^2 = 0$. *(2 marks)*

20. Complex Numbers, EXT2 N2 2010 HSC 2d

Let $z = \cos\theta + i\sin\theta$ where $0 < \theta < \dfrac{\pi}{2}$.

On the Argand diagram the point A represents z, the point B represents z^2 and the point C represents $z + z^2$.



Copy or trace the diagram into your writing booklet.

i. Explain why the parallelogram $OACB$ is a rhombus. ***(1 mark)***

ii. Show that $\arg\left(z + z^2\right) = \dfrac{3\theta}{2}$. ***(1 mark)***

iii. Show that $\left|z + z^2\right| = 2\cos\dfrac{\theta}{2}$. ***(2 marks)***

iv. By considering the real part of $z + z^2$, or otherwise deduce that

$$\cos\theta + \cos 2\theta = 2\cos\frac{\theta}{2}\cos\frac{3\theta}{2}.$$ ***(1 mark)***

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21. Complex Numbers, EXT2 N2 2011 HSC 4a

Let a and b be real numbers with $a \neq b$. Let $z = x + iy$ be a complex number such that

$$|\,z - a\,|^2 - |\,z - b\,|^2 = 1.$$

i. Prove that $x = \dfrac{a+b}{2} + \dfrac{1}{2(b-a)}$. ***(2 marks)***

ii. Hence, describe the locus of all complex numbers z such that $|\,z - a\,|^2 - |\,z - b\,|^2 = 1.$ ***(1 mark)***

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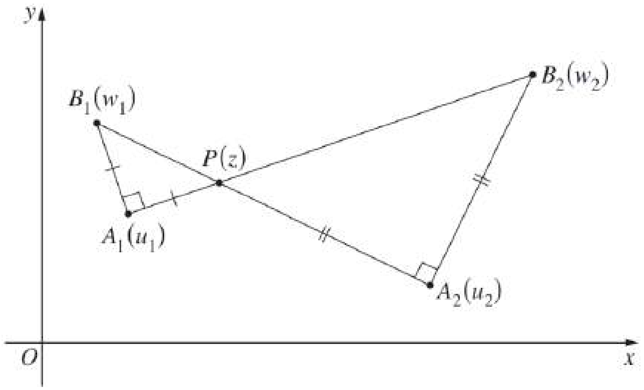
22. Complex Numbers, EXT2 N2 2012 HSC 12d

On the Argand diagram the points A_1 and A_2 correspond to the distinct complex numbers u_1 and u_2 respectively. Let P be a point corresponding to a third complex number z.

Points B_1 and B_2 are positioned so that ΔA_1PB_1 and ΔA_2B_2P, labelled in an anti-clockwise direction, are right-angled and isosceles with right angles at A_1 and A_2, respectively. The complex numbers w_1 and w_2 correspond to B_1 and B_2, respectively.



i. Explain why $w_1 = u_1 + i(z - u_1)$. ***(1 mark)***

ii. Find the locus of the midpoint of B_1B_2 as P varies. ***(2 marks)***

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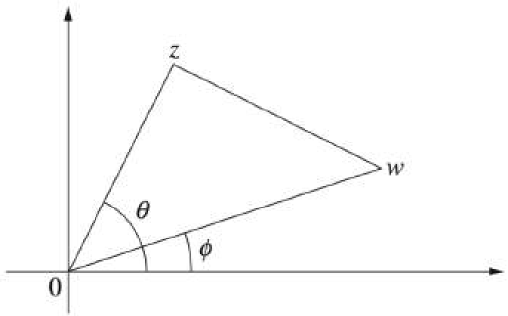
23. Complex Numbers, EXT2 N2 2013 HSC 15a

The Argand diagram shows complex numbers w and z with arguments ϕ and θ respectively, where $\phi < \theta$. The area of the triangle formed by $0, w$ and z is A.



Show that $z\overline{w} - w\overline{z} = 4iA.$ ***(3 marks)***

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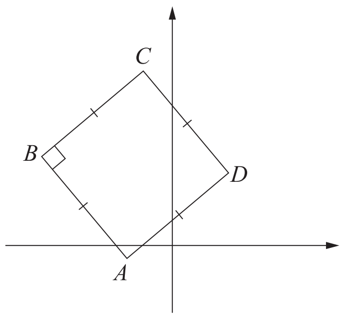
24. Complex Numbers, EXT2 N2 2017 HSC 13e

The points A, B, C and D on the Argand diagram represent the complex numbers a, b, c and d respectively. The points form a square as shown on the diagram.



By using vectors, or otherwise, show that $c = (1+i)d - ia$. *(2 marks)*

25. Complex Numbers, EXT2 N2 2016 HSC 16b

i. The complex numbers $0,\ u$ and v form the vertices of an equilateral triangle in the Argand diagram.

Show that $u^2 + v^2 = uv$. *(2 marks)*

ii. Give an example of non-zero complex numbers u and v, so that $0,\ u$ and v form the vertices of an equilateral triangle in the Argand diagram. *(1 mark)*

26. Complex Numbers, EXT2 N2 2019 HSC 16b

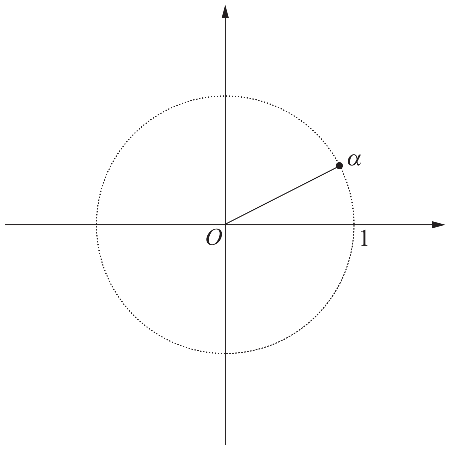
Let $P(z) = z^4 - 2kz^3 + 2k^2z^2 + mz + 1$, where k and m are real numbers.

The roots of $P(z)$ are $\alpha, \overline{\alpha}, \beta, \overline{\beta}$.

It is given that $|\alpha| = 1$ and $|\beta| = 1$.

i. Show that $(\mathrm{Re}(\alpha))^2 + (\mathrm{Re}(\beta))^2 = 1$. *(3 marks)*

ii. The diagram shows the position of α.



Copy or trace the diagram into your writing booklet.

On the diagram, accurately show all possible positions of β. *(2 marks)*

27. Complex Numbers, EXT2 N2 2011 HSC 6c

On an Argand diagram, sketch the region described by the inequality

$$\left|1 + \frac{1}{z}\right| \leq 1. \quad \textit{(2 marks)}$$

28. Complex Numbers, EXT2 N2 2016 HSC 16a

i. The complex numbers $z = \cos\theta + i\sin\theta$ and $w = \cos\alpha + i\sin\alpha$, where $-\pi < \theta < \pi$ and $-\pi < \alpha \leq \pi$, satisfy

$$1 + z + w = 0.$$

By considering the real and imaginary parts of $1 + z + w$, or otherwise, show that $1,\ z$ and w form the vertices of an equilateral triangle in the Argand diagram. *(3 marks)*

ii. Hence, or otherwise, show that if the three non-zero complex numbers $2i,\ z_1$ and z_2 satisfy

$$|\,2i\,| = |\,z_1\,| = |\,z_2\,| \text{ AND } 2i + z_1 + z_2 = 0.$$

then they form the vertices of an equilateral triangle in the Argand diagram. *(2 marks)*

Worked Solutions

1. Complex Numbers, EXT2 N2 2017 HSC 1 MC

$P(x)$ has 8 separate roots.

$\therefore$ Must be of degree at least 8.

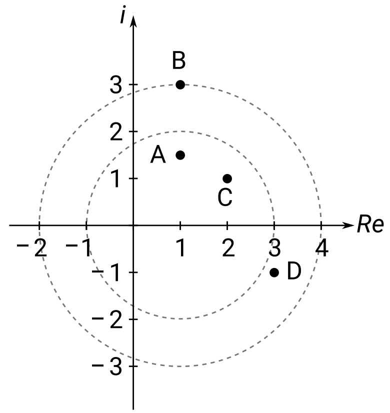
Since 1 is also a root,

$\Rightarrow\ C$

2. Complex Numbers, EXT2 N2 2017 HSC 3 MC

$|z-1| = 2$ is a circle with centre $(1,0)$, radius 2

$|z-1| = 3$ is a circle with centre $(1,0)$, radius 3



$\therefore$ Only $3 - i$ lies within the two circles,

i.e. satisfies $2 < |z-1| < 3$

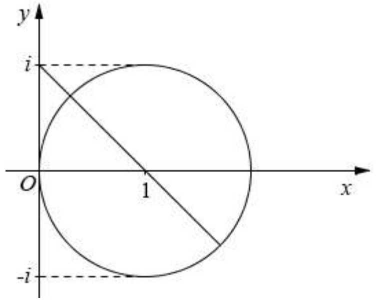
$\Rightarrow\ D$

3. Complex Numbers, EXT2 N2 2013 HSC 5 MC

$|z-1| = r$ is a circle with centre $(1,0)$, and radius r.

$\Rightarrow\ B$

4. Complex Numbers, EXT2 N2 2015 HSC 9 MC



$|z-1|=1$ is a circle, centre (1,0), radius 1.

Consider the graph:

Distance from $(0, i)$ to the centre $=\sqrt{2}$

$\therefore$ Greatest distance $=\sqrt{2}+\text{radius}=\sqrt{2}+1$

$\Rightarrow D$

5. Complex Numbers, EXT2 N2 2016 HSC 5 MC

$$\begin{aligned}\frac{1-i}{1+i} &= \frac{(1-i)^2}{(1+i)(1-i)}\\ &= \frac{-2i}{2}\\ &= -i\end{aligned}$$

$\therefore$ Clockwise rotation by $\dfrac{\pi}{2}$.

$\Rightarrow B$

6. Complex Numbers, EXT2 N2 2018 HSC 7 MC

$$\begin{aligned}\arg(z) &= \arg(z+1-i)\\ &= \arg(z-(-1+i))\end{aligned}$$

$\Rightarrow \arg(z-(-1+i))$ is the argument of z from $(-1+i)$.

Plot $(-1+i)$ on the argand diagram and then test different positions of z along the solutions for each option.

$\Rightarrow$ D

7. Complex Numbers, EXT2 N2 2011 HSC 2b

i. $$\begin{aligned} z &= 1+i\sqrt{3}+\sqrt{3}+i\\ &= (1+\sqrt{3})+i(1+\sqrt{3})\end{aligned}$$

ii. $$\arg z = \tan^{-1}\left(\frac{1+\sqrt{3}}{1+\sqrt{3}}\right)=\frac{\pi}{4}$$

$$\arg(\sqrt{3}+i)=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)=\frac{\pi}{6}$$

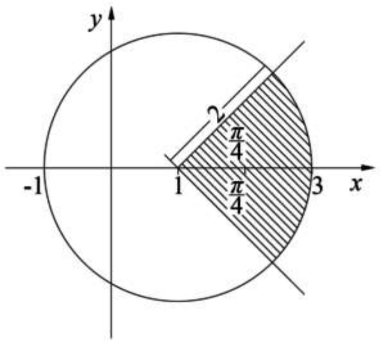
$$\text{Difference}=\frac{\pi}{4}-\frac{\pi}{6}=\frac{\pi}{12}$$

$\Rightarrow$ Opposite angles of a rhombus are equal

$\Rightarrow$ The diagonals of a rhombus bisect the angles

$$\begin{aligned}\therefore \theta &= \pi-2\times\frac{\pi}{12}\\ &= \frac{5\pi}{6} \quad \text{(angle sum of triangle)}\end{aligned}$$

8. Complex Numbers, EXT2 N2 2009 HSC 2d



9. Complex Numbers, EXT2 N2 2010 HSC 2c

Consider $\quad 1 \le |z| \le 2$

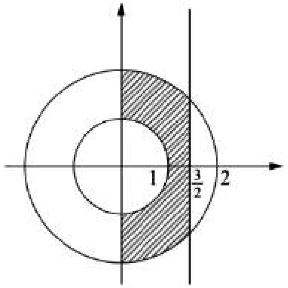
$$1 \le x^2 + y^2 \le 4$$

$z + \overline{z} = (x + iy) + (x - iy) = 2x$

Consider $\quad 0 \le z + \overline{z} \le 3$

$$0 \le 2x \le 3$$

$$0 \le x \le \frac{3}{2}$$

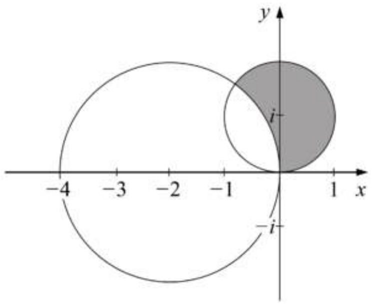


10. Complex Numbers, EXT2 N2 2012 HSC 11b

$|z+2| \ge 2$ and $|z - i| \le 1$

$(x+2)^2 + y^2 \ge 4$

$x^2 + (y-1)^2 \le 1$

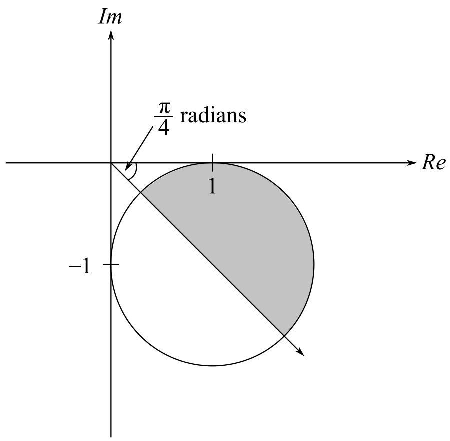


11. Complex Numbers, EXT2 N2 2017 HSC 11c

$|z - 1 + i| = 1$ is a circle with centre $(1, -1)$

and radius 1.

Shaded area: $-\dfrac{\pi}{4} \le \arg \le 0 \;\; \cap \;\; |z - 1 + i| \le 1$



12. Complex Numbers, EXT2 N2 2018 HSC 11d

i. $w = iu$

$$= i(5+2i)$$

$$= -2 + 5i$$

ii. $v = u + w$

$$= 5 + 2i + (-2 + 5i)$$

$$= 3 + 7i$$

iii. $\arg\left(\dfrac{w}{v}\right) = \arg(w) - \arg(v)$

$$= \frac{\pi}{4} \quad \text{(diagonal of square bisects corner)}$$

13. Complex Numbers, EXT2 N2 2007 HSC 2c

$$\begin{aligned}\frac{1}{z}+\frac{1}{\overline{z}} &= \frac{1}{x+iy}+\frac{1}{x-iy}\\ &= \frac{x-iy+x+iy}{x^2+y^2}\\ &= \frac{2x}{x^2+y^2}\end{aligned}$$

$$\begin{aligned}\frac{2x}{x^2+y^2} &= 1\\ x^2+y^2 &= 2x\\ x^2-2x+1+y^2 &= 1\\ (x-1)^2+y^2 &= 1\end{aligned}$$

$\therefore$ Locus is a circle, centre $(1,0)$, radius 1, excluding the point $(0,0)$.

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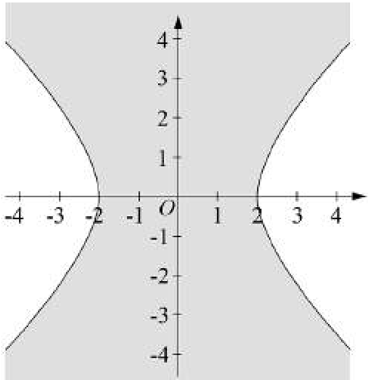
14. Complex Numbers, EXT2 N2 2013 HSC 11e

$$\begin{aligned}z^2+\overline{z}^2 &\le 8\\ (x+iy)^2+(x-iy)^2 &\le 8\\ x^2+2ixy-y^2+x^2-2ixy-y^2 &\le 8\\ 2x^2-2y^2 &\le 8\\ x^2-y^2 &\le 4\end{aligned}$$



15. Complex Numbers, EXT2 N2 2014 HSC 11c

Let $z = x + iy$

$$|\,x + iy\,| \le |\,x - 2 + iy\,|$$

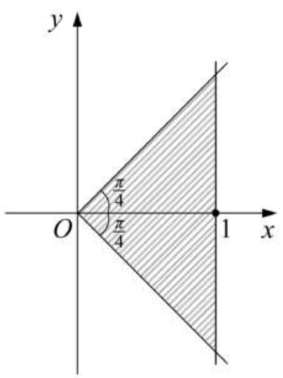
$$\sqrt{x^2 + y^2} \le \sqrt{(x-2)^2 + y^2}$$

$$x^2 + y^2 \le x^2 - 4x + 4 + y^2$$

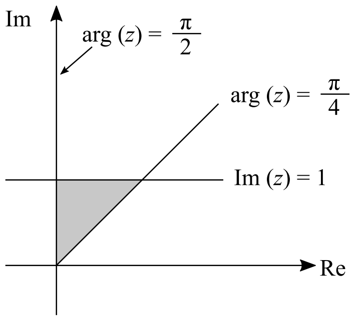
$$4x - 4 \le 0$$

$$x \le 1$$

$$-\frac{\pi}{4} \le \arg z \le \frac{\pi}{4}$$



16. Complex Numbers, EXT2 N2 2019 HSC 12a



Shaded Area: $\frac{\pi}{4} \le \arg(z) \le \frac{\pi}{2}$ **and** $\mathrm{Im}(z) \le 1$

17. Complex Numbers, EXT2 N2 2004 HSC 2c

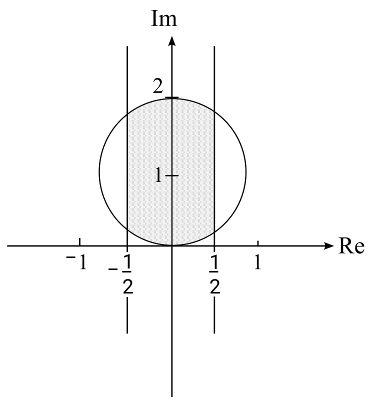
$$|z + \overline{z}| \le 1$$

$$|x + iy + x - iy| \le 1$$

$$|2x| \le 1$$

$$|x| \le \frac{1}{2}$$

$|z - i| \le 1 \;\Rightarrow\;$ **Circle, radius** $= 1$ **, centre** $(0, 1)$

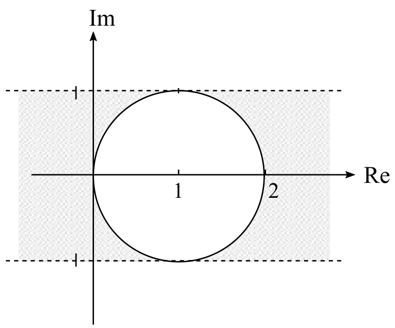


18. Complex Numbers, EXT2 N2 2005 HSC 2c

$$\begin{aligned} |z - \overline{z}| &< 2 \\ |x + iy - (x - iy)| &< 2 \\ |2iy| &< 2 \\ |y| &< 1 \end{aligned}$$

$|z - 1| = 1 \;\Rightarrow\;$ Circle, radius $= 1$, centre $(1, 0)$

$\therefore$ Graph: $|z - \overline{z}| < 2 \,\cap\, |z - 1| \geq 1$



19. Complex Numbers, EXT2 N2 2007 HSC 2d

i. Since ΔORQ is equilateral, each angle is $\dfrac{\pi}{3}$ radians.

From $R(a)$:

$Q(z_2)$ is an anticlockwise rotation through $\dfrac{\pi}{3}$.

$$\therefore\ z_2 = a\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = \omega a.$$

ii. Solution 1

Similarly, $a = z_1\omega$

$$\begin{aligned} z_1 &= \frac{a}{\omega} \\ : z_1 z_2 &= \frac{a}{\omega} \times \omega a \\ &= a^2 \end{aligned}$$

Solution 2

$P(z_1)$ is a clockwise rotation of $R(a)$ through $\dfrac{\pi}{3}$.

$$\begin{aligned} \therefore\ z_1 &= \overline{\omega}a. \\ \therefore\ z_1 z_2 &= \overline{\omega}a \times \omega a \\ &= a^2\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right) \times \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) \\ &= a^2\left(\cos^2\frac{\pi}{3} + \sin^2\frac{\pi}{3}\right) \\ &= a^2 \end{aligned}$$

iii. $z^2 - az + a^2 = 0$

Let the roots be α and β.

$$\alpha + \beta = -\frac{b}{a} = a$$

$$\alpha\beta = \frac{c}{a} = 2$$

$$\begin{aligned} z_1 z_2 &= a^2 \quad \text{(part (ii))} \\ z_1 + z_2 &= \overline{\omega}a + \omega a \\ &= \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3} + \cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)a \end{aligned}$$

$$= 2\cos\frac{\pi}{3} \times a$$

$$= 2 \times \frac{1}{2} \times a$$

$$= a$$

$\therefore\ z_1$ and z_2 are the roots of $z^2 - az + a^2 = 0.$

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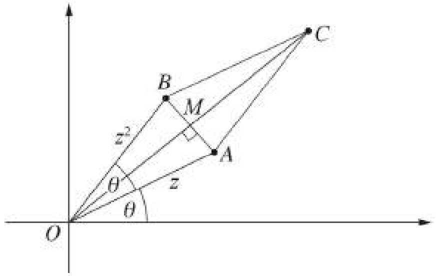
20. Complex Numbers, EXT2 N2 2010 HSC 2d

i. $z = \cos\theta + i\sin\theta$



$$|\,OA\,| = |\,z\,| = 1$$

$$|\,OB\,| = \left|\,z^2\,\right| = |\,z\,|^2 = 1$$

$\therefore\ OACB$ is a parallelogram with a pair of adjacent sides equal.

$\therefore\ OACB$ is a rhombus.

ii. $\arg z^2 = 2\arg z = 2\theta$

$$\angle BOA = 2\theta - \theta = \theta$$

Since $OACB$ is a rhombus then CO bisects $\angle BOA$

$$\therefore \angle COA = \frac{\theta}{2}$$

$$\therefore\ \arg\left(z + z^2\right) = \theta + \frac{\theta}{2} = \frac{3\theta}{2}$$

iii. $OC = \left|\,z + z^2\,\right|$

Join AB so that it meets OC at M

$AB \perp OC$, and $OM = OC$ (diagonals of a rhombus)

In ΔOAM:

♦♦ *Mean mark part (iii) 26%.*

$$\cos\frac{\theta}{2} = \frac{OM}{OA}$$

$$= OM$$

$$\therefore OC = 2 \times OM$$

$$= 2\cos\frac{\theta}{2}$$

iv. $z + z^2 = \cos\theta + i\sin\theta + (\cos\theta + i\sin\theta)^2$

$$= \cos\theta + \cos 2\theta + i(\sin\theta + \sin 2\theta) \quad \text{(De Moivre)}$$

$$\therefore \ \text{Re}\left(z + z^2\right) = \cos\theta + \cos 2\theta$$

Using parts (ii) and (iii),

♦♦ *Mean mark part (iv) 44%.*

$$z + z^2 = 2\cos\frac{\theta}{2}\left(\cos\frac{3\theta}{2} + i\sin\frac{3\theta}{2}\right)$$

Equating real parts:

$$\cos\theta + \cos 2\theta = 2\cos\frac{\theta}{2}\cos\frac{3\theta}{2}$$

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21. Complex Numbers, EXT2 N2 2011 HSC 4a

i.

$$\begin{aligned}
|z-a|^2 - |z-b|^2 &= 1\\
|(x-a)+iy|^2 - |(x-b)+iy|^2 &= 1\\
(x-a)^2 + y^2 - \left((x-b)^2 + y^2\right) &= 1\\
(x-a)^2 - (x-b)^2 &= 1\\
(x-a-(x-b))(x-a+x-b) &= 1\\
(b-a)(2x-a-b) &= 1
\end{aligned}$$

$$2x - a - b = \frac{1}{b-a}$$

$$2x = a + b + \frac{1}{b-a}$$

$$\therefore x = \frac{a+b}{2} + \frac{1}{2(b-a)}$$

ii. The locus is the vertical line

♦ *Mean mark part (ii) 44%.*

$$x = \frac{a+b}{2} + \frac{1}{2(b-a)}.$$

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22. Complex Numbers, EXT2 N2 2012 HSC 12d

i.

$$\overrightarrow{A_1P} = z - u_1$$

$$\overrightarrow{A_1B_1} = w_1 - u_1$$

$B_1A_1 \perp A_1P$ and $\left|\overrightarrow{A_1P}\right| = \left|\overrightarrow{A_1B_1}\right|$

$\overrightarrow{A_1B_1}$ is an anticlockwise rotation of $\overrightarrow{A_1P}$ through 90°

$$\therefore w_1 - u_1 = i(z - u_1)$$

$$\therefore w_1 = u_1 + i(z - u_1)$$

ii.

$$\overrightarrow{A_2B_2} = w_2 - u_2$$

$$\overrightarrow{A_2P} = z - u_2$$

$A_2B_2 \perp A_2P$ and $\left|\overrightarrow{A_2B_2}\right| = \left|\overrightarrow{A_2P}\right|$

$\overrightarrow{A_2P}$ is an anticlockwise rotation of $\overrightarrow{A_2B_2}$ through 90°

$$\begin{aligned}
z - u_2 &= i(w_2 - u_2)\\
iw_2 &= z - u_2 + iu_2\\
-w_2 &= iz - iu_2 - u_2\\
\therefore\ w_2 &= u_2 + i(u_2 - z)
\end{aligned}$$

$\therefore$ The midpoint of B_1B_2 is $\dfrac{w_1 + w_2}{2}$

$$\begin{aligned}
&= \frac{1}{2}[u_1 + i(z - u_1) + u_2 + i(u_2 - z)]\\
&= \frac{1}{2}[u_1 + u_2 + i(u_2 - u_1)]\\
&= \frac{u_1 + u_2}{2} + \frac{u_2 - u_1}{2}i \quad \text{(which is a fixed point)}
\end{aligned}$$

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23. Complex Numbers, EXT2 N2 2013 HSC 15a

♦♦ *Mean mark 29%.*
STRATEGY: The angles shown in the graphic should alert students that the mod-arg approach is likely to be easier than the $x+iy$ form.

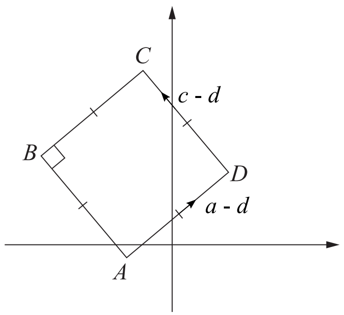
$$\begin{aligned} A &= \frac{1}{2}ab\sin C \\ &= \frac{1}{2}|z||w|\ \sin(\theta-\phi) \end{aligned}$$

$$\begin{aligned} z\overline{w} - w\overline{z} &= |z|\text{cis}\ \theta \cdot |w|\text{cis}(-\phi) - |w|\text{cis}\ \phi \cdot |z|\text{cis}(-\theta) \\ &= |z||w|(\text{cis}\ \theta\ \text{cis}(-\phi) - \text{cis}\ \phi\ \text{cis}(-\theta)) \\ &= |z||w|(\text{cis}(\theta-\phi) - \text{cis}(\phi-\theta)) \end{aligned}$$

$$\text{Since}\ \ \cos(\phi-\theta) = \cos(\theta-\phi), \quad \text{and}$$
$$\sin(\phi-\theta) = -\sin(\theta-\phi)$$
$$\Rightarrow \text{cis}(\theta-\phi) - \text{cis}(\phi-\theta) = 2i\sin(\theta-\phi)$$

$$\begin{aligned} \therefore z\overline{w} - w\overline{z} &= 2i|z||w|\sin(\theta-\phi) \\ &= 4i\left(\frac{1}{2}|z||w|\ \sin(\theta-\phi)\right) \\ &= 4iA \quad \text{... as required} \end{aligned}$$

24. Complex Numbers, EXT2 N2 2017 HSC 13e



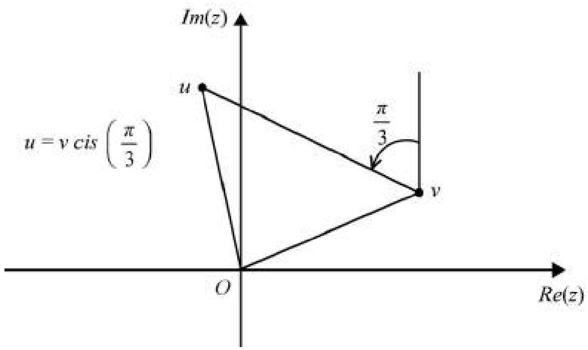
♦ *Mean mark 49%.*

$$c-d = i(d-a) \quad (\text{rotation of } \frac{\pi}{2})$$
$$\begin{aligned} \therefore c &= d + id - ia \\ &= (1+i)d - ia \ldots \text{as required.} \end{aligned}$$

25. Complex Numbers, EXT2 N2 2016 HSC 16b

i. $0, u, v$ are vertices of an equilateral triangle.

♦♦ *Mean mark 23%.*



$$\begin{aligned} \Rightarrow u &= v\text{cis}\left(\pm\frac{\pi}{3}\right) \\ u^3 &= v^3\text{cis}(\pm\pi) \\ u^3 &= -v^3 \\ u^3 + v^3 &= 0 \end{aligned}$$

$$(u+v)\left(u^2 - uv + v^2\right) = 0$$
Since $u \neq v$,
$$u^2 - uv + v^2 = 0$$
$$\therefore u^2 + v^2 = uv$$

ii. Let $v = 1$,

♦♦ *Mean mark 33%.*

$$\begin{aligned} \therefore u &= \text{cis}\left(\frac{\pi}{3}\right) \\ &= \cos\frac{\pi}{3} + i\sin\frac{\pi}{3} \\ &= \frac{1}{2} + \frac{\sqrt{3}}{2}i \end{aligned}$$

26. Complex Numbers, EXT2 N2 2019 HSC 16b

i. $P(z) = z^4 - 2kz^3 + 2k^2z^2 + mz + 1, \quad k, m \in \mathbb{R}$

Roots: $\alpha, \overline{\alpha}, \beta, \overline{\beta}$ and $|\alpha| = 1, |\beta| = 1$

Show $(\text{Re}(\alpha))^2 + (\text{Re}(\beta))^2 = 1$

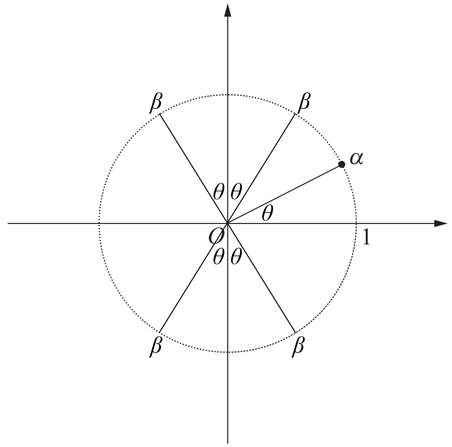
$$\alpha + \overline{\alpha} + \beta + \overline{\beta} = 2k$$
$$2\text{Re}(\alpha) + 2\text{Re}(\beta) = 2k$$
$$\text{Re}(\alpha) + \text{Re}(\beta) = k$$

$$\alpha\overline{\alpha} + \alpha\beta + \alpha\overline{\beta} + \overline{\alpha}\beta + \overline{\alpha}\overline{\beta} + \beta\overline{\beta} = 2k^2$$
$$|\alpha|^2 + \alpha(\beta + \overline{\beta}) + \overline{\alpha}(\beta + \overline{\beta}) + |\beta|^2 = 2k^2$$
$$1 + (\alpha + \overline{\alpha})(\beta + \overline{\beta}) + 1 = 2k^2$$
$$2 + 2\text{Re}(\alpha) \cdot 2\text{Re}(\beta) = 2(\text{Re}(\alpha) + \text{Re}(\beta))^2$$
$$2 + 4\text{Re}(\alpha)\text{Re}(\beta) = 2\text{Re}(\alpha)^2 + 4\text{Re}(\alpha)\text{Re}(\beta) + 2\text{Re}(\beta)^2$$
$$2 = 2\left(\text{Re}(\alpha)^2 + \text{Re}(\beta)^2\right)$$
$$\therefore 1 = \text{Re}(\alpha)^2 + \text{Re}(\beta)^2$$

♦♦ *Mean mark part (i) 26%.*

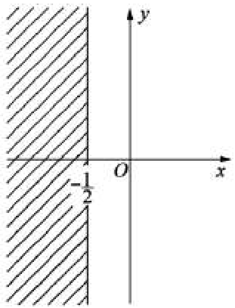
ii. $|\alpha| = |\beta|$ (given)

$$\text{Re}(\alpha)^2 + \text{Re}(\beta)^2 = 1 \quad \text{(see part (i))}$$
$$\text{Re}(\alpha)^2 + \text{Im}(\alpha)^2 = 1 \quad (|\alpha| = 1)$$
$$\Rightarrow \text{Re}(\beta)^2 = \text{Im}(\alpha)^2$$
$$\text{Re}(\beta) = \pm\text{Im}(\alpha)$$



♦♦♦ *Mean mark part (ii) 10%.*

27. Complex Numbers, EXT2 N2 2011 HSC 6c



$$\left|1 + \frac{1}{z}\right| \leq 1$$
$$\left|\frac{z+1}{z}\right| \leq 1$$
$$\frac{|z+1|}{|z|} \leq 1$$
$$|z+1| \leq |z|$$
$$\sqrt{(x+1)^2 + y^2} \leq \sqrt{x^2 + y^2}$$
$$(x+1)^2 + y^2 \leq x^2 + y^2$$
$$x^2 + 2x + 1 \leq x^2$$
$$\therefore x \leq -\frac{1}{2}$$

♦♦♦ *Mean mark 18%.*
MARKER'S COMMENT: Substituting $z = x + iy$ immediately was common and caused major algebraic problems.

28. Complex Numbers, EXT2 N2 2016 HSC 16a

i. $z = \cos\theta + i\sin\theta, \ |z| = 1$

$w = \cos\alpha + i\sin\alpha, \ |w| = 1$

Since $1 + z + w = 0$

$$\text{Im}(1+z+w) = 0$$

$$\sin\theta + \sin\alpha = 0$$

$$\sin\theta = -\sin\alpha$$

$$\therefore \theta = -\alpha \ldots (1)$$

$$\text{Re}(1+z+w) = 0$$

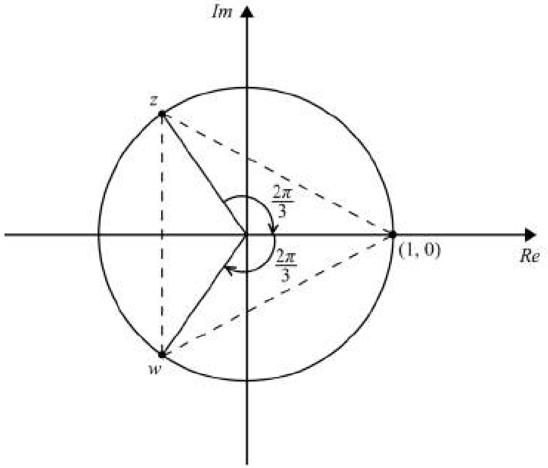
$$1 + \cos\theta + \cos\alpha = 0$$

$$2\cos\theta = -1 \quad (\text{Since } \cos\alpha = \cos(-\theta) = \cos\theta)$$

$$\cos\theta = -\frac{1}{2}$$

$$\therefore \theta = \frac{2\pi}{3}$$

$$\alpha = -\frac{2\pi}{3}$$



$\Rightarrow$ All points are on the unit circle separated by $\dfrac{2\pi}{3}$ radians.

♦♦♦ *Mean mark 25%.*
STRATEGY: A clear graphical image simplifies both parts of this question significantly.

$\therefore$ They are vertices of an equilateral triangle.

ii. $|2i| = 2$

Let $z_1 = 2(\cos\theta + i\sin\theta), \ |z_1| = 2$

$z_2 = 2(\cos\alpha + i\sin\alpha), \ |z_2| = 2$

$$\text{Re}(2i + z_1 + z_2) = 0$$

$$2(\cos\theta + \cos\alpha) = 0$$

$$\therefore \cos\theta = -\cos\alpha$$

$$\therefore \theta = \pi - \alpha$$

$$\text{Im}(2i + z_1 + z_2) = 0$$

$$2(1 + \sin\theta + \sin\alpha) = 0$$

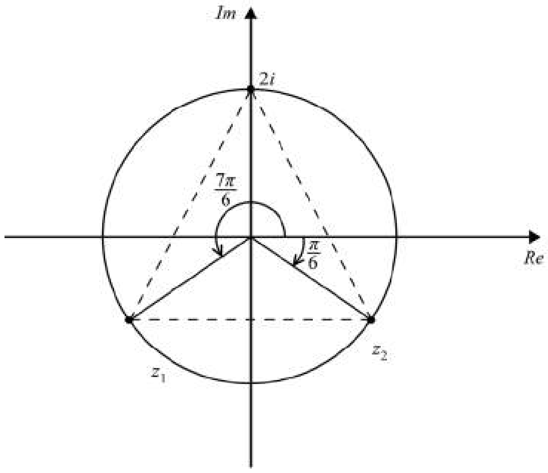
$$\sin\theta + \sin\alpha = -1$$

$$2\sin\theta = -1 \quad (\text{Since } \sin\alpha = \sin(\pi - \theta) = \sin\theta)$$

$$\sin\theta = -\frac{1}{2}$$

$$\therefore \theta = \frac{7\pi}{6}$$

$$\therefore \alpha = -\frac{\pi}{6}$$



$\Rightarrow$ All points are on the 2 unit

♦♦♦ *Mean mark 5%.*

circle separated by $\frac{2\pi}{3}$ radians.

$\therefore$ $2i$, z_1 and z_2 are vertices of an equilateral triangle.